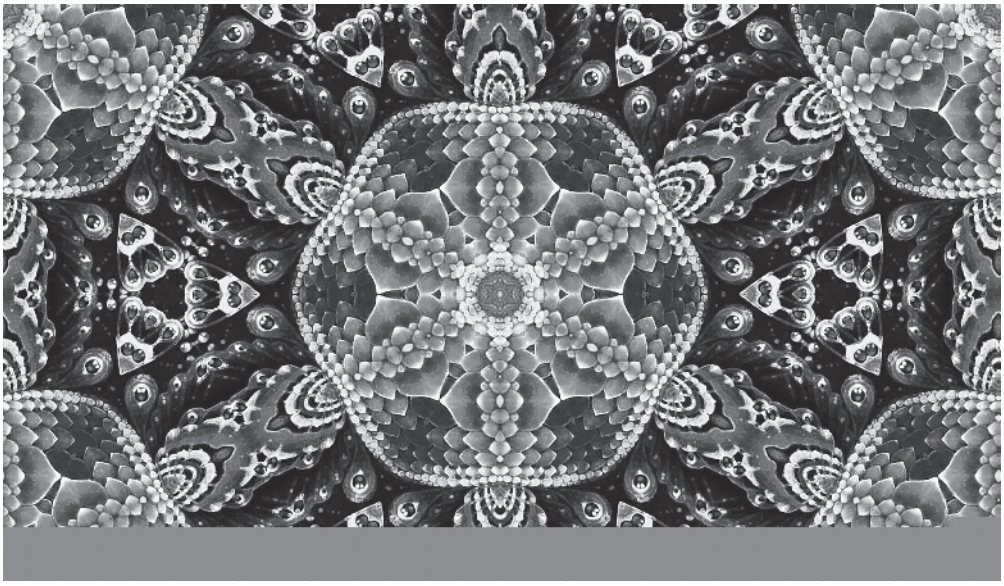


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4.25 Figure

EXERCISES 4

Computation

In Exercises 1 through 5, compute the indicated product involving the following permutations in S_6:

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 1 & 4 & 5 & 6 & 2 \end{pmatrix}, \qquad \tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 1 & 3 & 6 & 5 \end{pmatrix}, \qquad \mu = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 2 & 4 & 3 & 1 & 6 \end{pmatrix}.$$

1. $\tau\sigma$ **2.** $\tau^2\sigma$ **3.** $\mu\sigma^2$ **4.** $\sigma^{-2}\tau$ **5.** $\sigma^{-1}\tau\sigma$

In Exercises 6 through 9, compute the expressions shown for the permutations σ, τ, and μ defined prior to Exercise 1.

6. σ^6 **7.** μ^2 **8.** σ^{100} **9.** μ^{100}

10. Convert the permutations σ, τ, and μ defined prior to Exercise 1 to disjoint cycle notation.

11. Convert the following permutations in S_8 from disjoint cycle notation to two-row notation.

a. $(1, 4, 5)(2, 3)$
b. $(1, 8, 5)(2, 6, 7, 3, 4)$
c. $(1, 2, 3)(4, 5)(6, 7, 8)$

12. Compute the permutation products.

a. $(1, 5, 2, 4)(1, 5, 2, 3)$
b. $(1, 5, 3)(1, 2, 3, 4, 5, 6)(1, 5, 3)^{-1}$
c. $[(1, 6, 7, 2)^2(4, 5, 2, 6)^{-1}(1, 7, 3)]^{-1}$
d. $(1, 6)(1, 5)(1, 4)(1, 3)(1, 2)$

13. Compute the following elements of D_{12}. Write your answer in standard form.

a. $\mu\rho^2\mu\rho^8$
b. $\mu\rho^{10}\mu\rho^{-1}$
c. $\rho\mu\rho^{-1}$
d. $(\mu\rho^3\mu^{-1}\rho^{-1})^{-1}$

14. Write the group table for D_3. Compare the group tables for D_3 and S_3. Are the groups isomorphic?

Let A be a set and let $\sigma \in S_A$. For a fixed $a \in A$, the set

$$\mathcal{O}_{a,\sigma} = \{\sigma^n(a) \mid n \in \mathbb{Z}\}$$

is the **orbit** of a **under** σ. In Exercises 15 through 17, find the orbit of 1 under the permutation defined prior to Exercise 1.

15. σ **16.** τ **17.** μ

18. Verify that $H = \{\iota, \mu, \rho^2, \mu\rho^2\} \subseteq D_4$ is a group using the operation function composition.

19. **a.** Verify that the six matrices

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

form a group under matrix multiplication. [*Hint*: Don't try to compute all products of these matrices. Instead, think how the column vector $\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$ is transformed by multiplying it on the left by each of the matrices.]

b. What group discussed in this section is isomorphic to this group of six matrices?

20. After working Exercise 18, write down eight matrices that form a group under matrix multiplication that is isomorphic to D_4.

Concepts

In Exercises 21 through 23, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

21. The *dihedral group* D_n is the set of all functions $\phi : \mathbb{Z}_n \to \mathbb{Z}_n$ such that the line segment between vertex i and vertex j of U_n is an edge of P_n if and only if the line segment between vertices $\phi(i)$ and $\phi(j)$ in U_n is an edge of P_n.

22. A *permutation* of a set S is a one-to-one map from S to S.

23. The *order* of a group is the number of elements in the group.

In Exercises 24 through 28, determine whether the given function is a permutation of $\mathbb{R}$.

24. $f_1 : \mathbb{R} \to \mathbb{R}$ defined by $f_1(x) = x + 1$

25. $f_2 : \mathbb{R} \to \mathbb{R}$ defined by $f_2(x) = x^2$

26. $f_3 : \mathbb{R} \to \mathbb{R}$ defined by $f_3(x) = -x^3$

27. $f_4 : \mathbb{R} \to \mathbb{R}$ defined by $f_4(x) = e^x$

28. $f_5 : \mathbb{R} \to \mathbb{R}$ defined by $f_5(x) = x^3 - x^2 - 2x$

29. Determine whether each of the following is true or false.

a. Every permutation is a one-to-one function.

b. Every function is a permutation if and only if it is one-to-one.

c. Every function from a finite set onto itself must be one-to-one.

d. Every subset of an abelian group G that is also a group using the same operation as G is abelian.

e. The symmetric group S_{10} has 10 elements.

f. If $\phi \in D_n$, then ϕ is a permutation on the set $\mathbb{Z}_n$.

g. The group D_n has exactly n elements.

h. D_3 is a subset of D_4.

Theory

30. Let $n \geq 3$ and $k \in \mathbb{Z}_n$. Prove that in D_n, $\rho^k\mu = \mu\rho^{n-k}$.

31. Show that S_n is a nonabelian group for $n \geq 3$.